BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 304 (July 2022) **C1**
Digital Electronics Laboratory

Final Project Report

An Approach To Make a Basic Mechanical Braille Cell

Ethics Statement:

IMPORTANT! Please carefully read and sign the Ethics Statement, below. <u>Type the student ID and Write your name in your own handwriting</u>. *You will not receive credit for this project unless this statement is signed in the presence of your lab instructor.*

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), or copied from internet. We have cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Full Name: Akil Ahmad Taki **Student ID:**1806139	**Full Name:**Md. Jawadul Karim **Student ID:**1806162
Full Name: Syed Muid Al Kais **Student ID:**1806158	**Full Name:** Muhiul Islam Muhiuddin **Student ID:1806163**
Full Name: Khondokar Zahin **Student ID:**1806161	

Evaluation Form:

STEP	Assessment Tool	Criteria	CO	PO	MAX	SCORE
1	Peer Assessment	Individual Contribution	CO5	PO9	10	
2		Teamwork	CO5	PO9	10	
3		Ethics	CO4	PO8	10	
4	Viva	Ethics	CO4	PO8	10	
5		Tool Usage	CO2	PO5	10	
6	Report	Technological Limit Evaluation	CO2	PO5	10	
7		Technical Details	CO6	PO10	10	
8		Design Considerations	CO3	PO3	10	
9	Project Demonstration		CO3	PO3	10	
10	Recorded Video Presentation		CO6	PO10	10	
	TOTAL				100	

Course Instructor:

1. **Nafis Sadik**
2. **Shafayeth Jamil**

Signature of Evaluator: ______________________________

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1 Abstract

This braille design project aims to create an inclusive and accessible learning environment for visually impaired students. The project will involve designing braille materials for various subjects, including math, science, and language arts. The materials will be created using eco-friendly materials, and the designs will promote efficient use of space while maintaining legibility and readability. The project will also involve developing braille signage for public spaces such as elevators, restrooms, and classrooms to promote inclusivity and accessibility. The final design will be evaluated for its effectiveness in promoting accessibility and inclusivity and will be shared with schools and organizations to encourage the use of inclusive design practices. Overall, this project seeks to create a more accessible and inclusive learning environment for visually impaired students while promoting sustainability and environmental responsibility.

2 Introduction

A refreshable mechanical braille display is an electronic device that allows blind and visually impaired individuals to read text in braille by raising and lowering a series of pins. It consists of a surface with a grid of small cells, each of which contains one or more pins that can be raised or lowered to create braille characters.

When a user inputs text into the device, such as by connecting it to a computer or the device converts the text into braille and displays it by raising and lowering the appropriate pins. The user can then read the text by running their fingers over the pins, feeling the raised bumps that represent each letter, number, or symbol.

Refreshable mechanical braille displays are an important tool for blind and visually impaired individuals, as they allow for the quick and easy reading of digital text without the need for bulky printed materials. They are also versatile, as they can be used with a variety of devices and can display multiple languages and character sets.

2.1Complexity Analysis:

The design of a braille display is a complex problem because it involves multiple factors that need to be balanced and optimized to create an effective solution. Some of these factors include:

Size and portability: Braille displays need to be compact and portable enough to be carried around, but also large enough to accommodate enough braille cells to display readable text.

Durability: Braille displays need to be durable and able to withstand constant use and wear and tear.

Affordability: Braille displays need to be affordable enough to be accessible to the people who need them, while also meeting high quality standards.

User experience: Braille displays need to be easy to use and provide a comfortable, intuitive reading experience for the user.

Compatibility: Braille displays need to be compatible with a wide range of devices and software platforms.

These factors can lead to conflicting solutions that need to be balanced in order to create an optimal design. For example, a more affordable device might sacrifice durability or user experience.

In addition to these factors, the design of a braille display is also complex because it involves the integration of multiple engineering disciplines, such as mechanical, electrical, and software engineering. These disciplines need to work together to create a cohesive, functional device that meets the needs of the user.

Overall, the design of a braille display is a complex engineering problem that requires careful consideration of multiple factors and integration of multiple engineering disciplines. The optimal solution might not be immediately obvious and may require significant research, testing, and iteration to achieve.

3 Technical Details of the Design :

Here our design is just a prototype design. However, some common technical details that are important to consider in the development of a prototype braille design include:

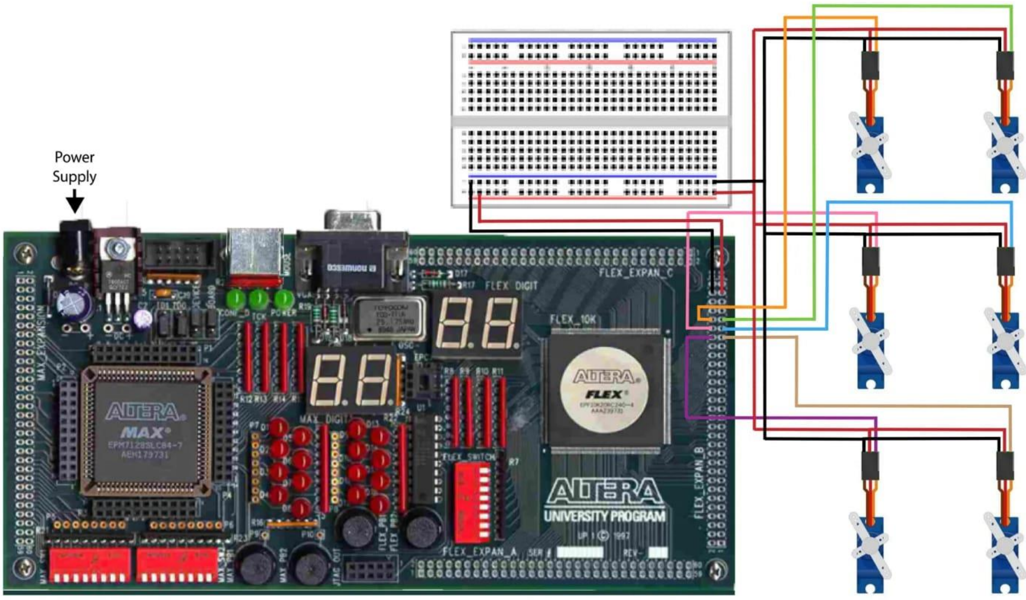
Braille cell design: The braille cell design will determine the size, spacing, and materials used to create the raised dots that form the braille characters. Here in our prototype we just use a cardboard as our cell.

Actuation mechanism: The actuation mechanism will raise and lower the pins in the braille cells. For our prototype we use a stepper motor, might be used to demonstrate the basic concept.

Control electronics: The control electronics will translate digital text into braille and send signals to the actuation mechanism. For a prototype,we use FPGA board to demonstrate the basic functionality.

Power source: The power source will be required to power the control electronics and actuation mechanism. For a prototype, a small battery or USB power supply might be used.

3.1 Design Method :



3.2Novelty Statement:

Here in our project, we use FPGA board where there is no USB port. But there is ps2 port .We used keyboard to take input through ps2 port .It converts letter to scan code .That's why we need to convert the scan code to ASCII code as we already made logical expression by using ASCII code .By this logic, we control our servo motor and verify our response with input.

3.3Pictures of Final Implementation

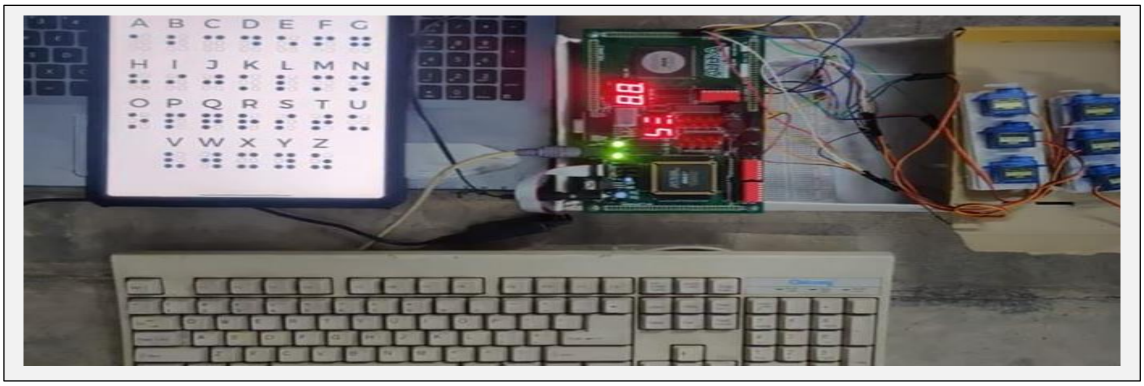


Figure 1: Implementation of Braille Design

3.4 YouTube Link:

Here we are attaching drive link instead of YouTube. We hope that you will consider this.

https://drive.google.com/file/d/1Cg6_M9VCYYHz9o0PkQrobk8jLy_o1wvX/view?fbclid=IwAR0FMYl4UKYdbORxXDDDZoFPi6cdmVXbrphgBnXU8d9Uv7mWDBTZt-ujm1k

4 Practical Design Considerations

4.1 Considerations to Public Health and Safety

The development of the braille system was primarily driven by the need to provide people with visual impairments the ability to read and write independently. However, the design of the braille system also took into consideration public health and safety concerns.

One way in which braille design addresses public health is by promoting hygiene. In the early years of the development of the braille system, raised dots were created by using materials like nails or other sharp objects that could easily cause injury and infection. In response to this, the braille design was improved to use safer materials like styluses that were rounded and did not have sharp edges. This helped to reduce the risk of infection and injury to individuals who used the system.

Another way in which the braille system addresses public health and safety is through its tactile nature. The tactile nature of the braille system allows individuals with visual impairments to read and write without relying on visual cues. This is particularly important in situations where there may be low lighting or other environmental factors that could pose a risk to public safety. For example, a person with a visual impairment can use the braille system to read important safety information like exit signs in a dimly lit area.

Furthermore, the braille system is widely used in public spaces such as elevators, train stations, airports, and other public facilities to provide information to individuals with visual impairments. This not only promotes accessibility but also promotes safety by ensuring that everyone can access important information in case of an emergency.

In summary, the design of the braille system was done with due consideration to public health and safety concerns. The use of safer materials for creating raised dots, the tactile nature of the system, and its wide application in public spaces are all examples of how braille design addresses these concerns.

4.2 Considerations to Environment

The design of the braille system was primarily focused on addressing the needs of people with visual impairments. However, the braille system was also developed with consideration for the environment.

One way in which the braille system is environmentally friendly is by promoting the use of recycled materials. The materials used to create braille documents, such as paper, can be recycled just like any other paper product. This promotes the reduction of waste and the conservation of natural resources.

In addition to promoting the use of recycled materials, the braille system also promotes efficient use of space. Because braille characters are raised dots, they take up very little space on a page compared to traditional printed text. This means that more information can be printed on a single page, reducing the amount of paper required to print a given amount of information. This not only reduces waste but also saves money on paper and printing costs.

Furthermore, the use of braille in public spaces promotes environmental sustainability by reducing the need for electronic devices and displays. For example, braille is commonly used on elevator buttons, signage, and other public displays to provide information to people with visual impairments. By using braille, these facilities can provide information without relying on electronic displays, which require energy to operate and can be less reliable in emergency situations.

In summary, the design of the braille system takes into consideration the environment by promoting the use of recycled materials, efficient use of space, and reducing the need for electronic devices and displays in public spaces. These design features contribute to environmental sustainability and conservation of natural resources.

4.3Considerations to Cultural, and Societal Needs

The development of the braille system was primarily driven by the need to provide people with visual impairments the ability to read and write independently. However, the design of the braille system also took into consideration cultural and societal needs.

One way in which the braille system addresses cultural and societal needs is by promoting accessibility and inclusivity. By providing a means for people with visual impairments to read and write, the braille system promotes equal access to education, employment, and other opportunities. This helps to break down barriers and promote social inclusion for people with visual impairments.

Another way in which the braille system addresses cultural and societal needs is through its versatility. The braille system can be used to write in any language, including languages that do not use the Roman alphabet. This makes it a valuable tool for promoting literacy and education in diverse communities around the world. Additionally, the braille system can be used to transcribe music, mathematics, and other specialized fields, making it a versatile tool for communication and expression.

Furthermore, the use of braille in public spaces promotes cultural awareness and sensitivity. By providing braille signage, menus, and other displays, facilities can demonstrate their commitment to inclusivity and accessibility. This not only promotes social inclusion but also demonstrates respect for people with visual impairments and their cultural and societal needs.

In summary, the design of the braille system takes into consideration cultural and societal needs by promoting accessibility, inclusivity, versatility, and cultural sensitivity. These design features contribute to breaking down barriers and promoting social inclusion for people with visual impairments.

5 Reflection on Individual and Team work

5.1 Individual Contribution of Each Member

In our project group we are 5 members. Each member of my group are so cooperative and did their assigned work perfectly.
As in our project we first need to convert input ASCII code to logical expression by K-mapping. In this task we all took 5/6 letter to find it's logical expression.
Then two of us handle the software part in Quartus and rest three of us took hardware part like controlling the servo motor and prototype design

5.2Diversity Statement of Team

- From the beginning my team bring different perspectives and ideas, leading to more innovative and creative solutions.
- They can approach a problem from different angles and develop more comprehensive solutions.
- They can consider multiple perspectives and make informed decisions that consider the needs and perspectives of all stakeholders.

6 References and Acknowledgement

6.1 Acknowledgement

We would like to express my sincere gratitude to all those who have contributed to the completion of this project.

First and foremost, we would like to thank our course teacher, for his invaluable guidance, support, and encouragement throughout the project. His insights and expertise have been instrumental in shaping this project and ensuring its quality.

I would also like to thank the members of my team for their diligent efforts and cooperation. Their hard work and dedication have been pivotal in achieving the objectives of this project.

We grateful to Nafis Sadik, Sir & Shafayeth Jamil, Sir for providing us FPGA board from laboratory which was crucial to the success of this project.

Once again, thank you to everyone who has contributed to this project in one way or another. Their assistance and support are deeply appreciated.

6.2 References

http://users.utcluj.ro/~baruch/sie/labor/PS2/Scan_Codes_Set_2.htm
https://braillebug.org/braille/braille-deciphering/
http://sticksandstones.kstrom.com/appen.html